

rAge

Assume that the youngest student is 10 years old. The age of the next older student can be computed by adding 2 years to the age of the previous younger student. The students are arranged in ascending order according to their age with the youngest student as the first one. Write a **recursive** function `rAge()` that takes in the rank of a student `studRank` and returns the age of the student to the calling function. For example, if `studRank` is 4, then the age of the corresponding student 16 will be returned. The recursive function is given below:

```
rAge(n) = 10           if n = 1
rAge(n) = rAge(n-1)+2  if n > 1
```

The function prototype is given as follows:

```
int rAge(int studRank);
```

A sample program template is given below to test the function:

```
#include <stdio.h>
int rAge(int studRank);
int main()
{
    int studRank;

    printf("Enter student rank: \n");
    scanf("%d",&studRank);
    printf("rAge(): %d\n", rAge(studRank));
    return 0;
}
int rAge(int studRank)
{
    /* Write your code here */
}
```

Some sample input and output sessions are given below:

(1) Test Case 1:
Enter student rank:
5
rAge(): 18

(2) Test Case 2:
Enter student rank:
1
rAge(): 10